The Effect of Digital Infrastructure on Household Income

Alice Smith and Robert Brown
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Abstract
This study examines whether broadband infrastructure changes household income. We use a difference-in-differences research design with county-level panel data.

Data and Sample

We construct a balanced panel of 500 counties from 2010 to 2020. The final sample contains 5,500 county-year observations. Income data come from the public household survey and broadband data come from the regulator annual reports.

Empirical Method

Our baseline model is a two-way fixed effects estimator with county and year fixed effects. Standard errors are clustered by county. We conduct event-study tests, placebo timing tests, and robustness checks using alternative income measures.

Results and Conclusion

Broadband access increases median household income by 4.2 percent. The effects are larger in rural counties. The parallel trends test supports the identification assumption. The study is limited by county-level aggregation and potential unobserved local policies.